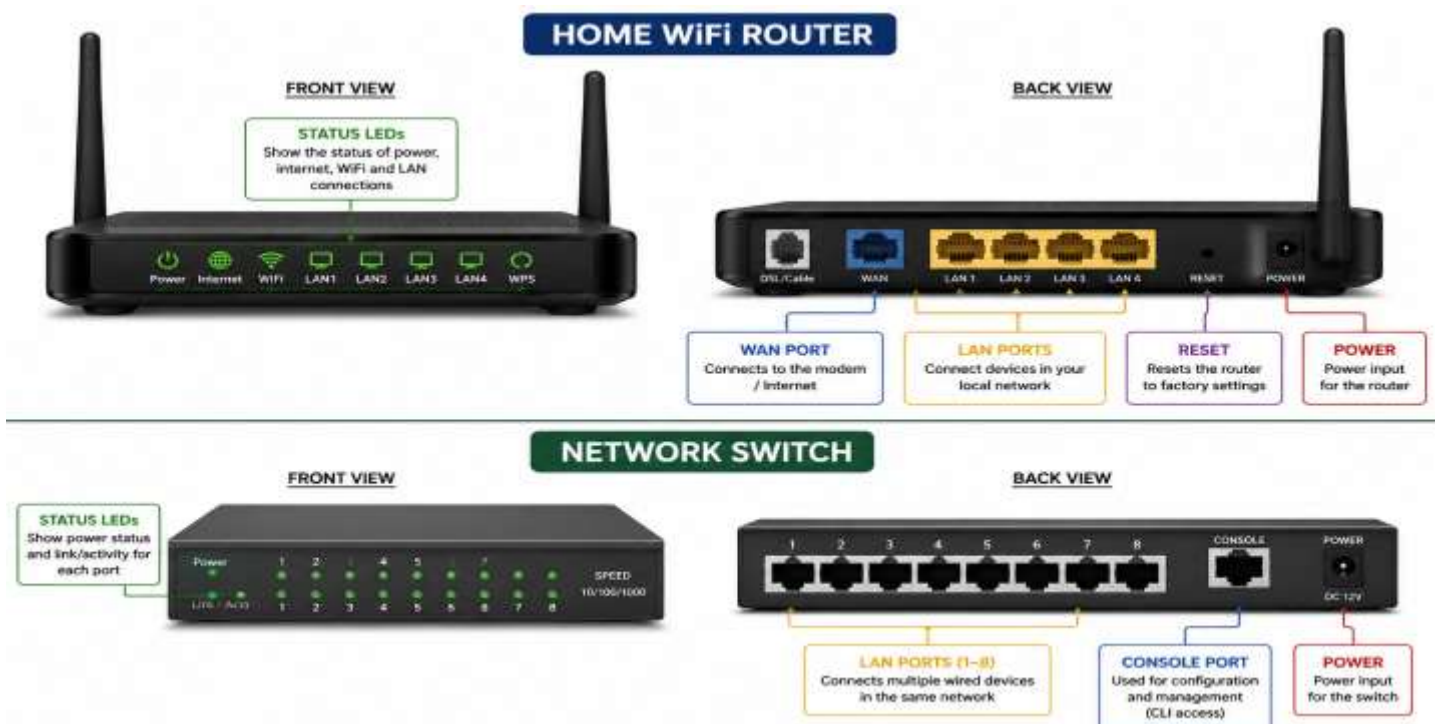


Comptia A+ Assignment

A+ - Understanding and maintenance of Networks

1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

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Home Wi-Fi Router

1. Status LEDs – Power, Internet, Wi-Fi aur LAN connection ki status dikhati hain.
2. WAN Port – Modem/Internet se router ko connect karta hai.
3. LAN Ports – Computer, printer jaise wired devices ko connect karte hain.
4. Reset Button – Router ko factory settings par reset karta hai.
5. Power Port – Router ko electricity supply deta hai.

Network Switch

1. Status LEDs – Power aur har port ki connection/activity dikhati hain.
2. LAN Ports – Multiple computers aur other wired devices ko same network mein connect karte hain.

3. Console Port – Network administrator switch ko configure/manage karne ke liye use karta hai.
4. Power Port – Switch ko power supply deta hai.

2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

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Difference	Router	Switch
1. Main function	Connects different networks together, such as a home network to the Internet.	Connects devices within the same network, such as PCs, printers, and servers.
2. Addresses used	Mainly uses IP addresses to decide where network packets should go.	Mainly uses MAC addresses to forward Ethernet frames to the correct device.
3. Traffic handling & networks	Routes traffic between different IP networks/subnets and can connect a LAN to the Internet/WAN.	Forwards traffic within a LAN and sends data only to the appropriate connected port when it knows the destination MAC address.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

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1. Wi-Fi Router

Part	Position	Purpose
Power Port	Usually on the back	Connects the router to its power adapter.
WAN/Internet Port	Back, often separated or differently colored	Connects the router to the modem/ISP for Internet access.
LAN Ethernet Ports	Back, usually 2–4 ports	Connect computers, TVs, printers, or other wired devices to the router.
LED Indicators	Usually on the front/top	Show power, Internet/WAN status, Wi-Fi activity, and LAN activity.

2. Network Switch

Part	Position	Purpose
Power Port	Usually at the back	Connects the switch to electrical power.
Ethernet Port 1	Front or back	Connects one network device using an Ethernet cable.
Ethernet Port 2	Next to Port 1	Connects another network device.
LED Indicators	Usually beside/above each Ethernet port	Show whether a connection is active and sometimes indicate connection speed/activity.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

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- **Small Gaming Café Network Scenario**

For a small gaming café with 8 gaming PCs, I would connect the router first to the ISP line. The router receives the Internet connection from the ISP and uses IP addresses to manage traffic between the café's local network and the Internet.

Then, I would connect one of the router's LAN ports to a network switch using an Ethernet cable. The 8 gaming PCs would each connect to a separate Ethernet port on the switch.

- **Network layout:**

ISP → Router → Switch → 8 Gaming PCs

I would use a switch because it provides multiple Ethernet ports and efficiently forwards data between the gaming PCs using their MAC addresses. The router provides Internet access, while the switch allows all 8 PCs to communicate with the router and each other. This setup is suitable for IPL streaming and multiplayer gaming because wired Ethernet connections provide a stable and reliable connection with low latency.

